

Data Session 4 Key, part 1: A Century and a Half of Campaigns

84-355 Democracy's Data | Session 4 — Returns over time (self-study) | Instructor copy — do not distribute

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All Part 7 code is given to students. Grade the interpretation, not the code.
Every number below was run against the data file.

The lab in one sentence

American campaigns transformed partly because the map they ran on transformed — a country where a third of the states were uncontested for eighty years became one where a dozen states decide everything, and that shift explains more about modern campaign behaviour than any change in technique.

The three findings students should leave with

```
a <- gap_for(c11)
a[a$year %in% c(1924, 1948, 1964, 1976, 1980, 1984, 2024), c("year", "south", "rest", "gap")]
```

```
#>   year    south    rest    gap
#> 16 1924 74.02989 32.30278 41.7
#> 22 1948 68.97906 51.35009 17.6
#> 26 1964 47.93240 63.08744 -15.2
#> 29 1976 55.89519 49.07220  6.8
#> 30 1980 48.00967 42.47186  5.5
#> 31 1984 37.95374 40.21913 -2.3
#> 41 2024 41.69793 49.10741 -7.4
```

```
sp <- aggregate(d2 ~ year, data = pres, FUN = sd, na.rm = TRUE); names(sp)[2] <- "sd"
sp[sp$year %in% c(1924, 1960, 2024), ]
```

```
#>   year      sd
#> 16 1924 21.407530
#> 25 1960  5.825148
#> 41 2024 12.052569
```

```
pres$close <- abs(pres$d2 - 50) < 5
nc <- aggregate(close ~ year, data = pres, FUN = sum, na.rm = TRUE)
nc[nc$year %in% c(1960, 1976, 2024), ]
```

```
#>   year close
#> 25 1960    33
#> 29 1976    31
#> 41 2024    13
```

1. **The South flips.** Gap peaks at **+41.7 in 1924**, is **+17.6 in 1948**, goes negative for the first time in **1964**, returns positive under Carter (**+6.8 in 1976**, **+5.5 in 1980**), and is negative in **every election from 1984 onward**, reaching **−7.4 in 2024**.
2. **State-to-state spread is U-shaped.** SD bottoms at **5.83 in 1960** and **roughly 2.1× that by 2024, at 12.05** — but for a different reason than the peak of 21.4 in 1924. In 1924, **71.8%** of the state-to-state variation was the South differing from everywhere else; in 2024 that is **6.5%**. The country is spread out again, but along different seams.
3. **The battleground shrinks.** **33** states within 5 points in 1960; **13** in 2024, out of 51 states rather than the 50 that existed then.

Option A — Move the line

Verified output

```
b <- gap_for(c17)
m <- merge(a[, c("year", "gap")], b[, c("year", "gap")], by = "year",
           suffixes = c("_11", "_17"))
m[m$year %in% c(1924, 1948, 1960, 1964, 1976, 1984, 2024), ]
```

```
#>   year gap_11 gap_17
#> 16 1924   41.7   35.4
#> 22 1948   17.6   13.6
#> 25 1960    5.8    4.3
#> 26 1964  -15.2   -9.5
#> 29 1976    6.8    6.6
#> 31 1984   -2.3   -1.4
#> 41 2024   -7.4   -8.0
```

The magnitudes shrink; the story does not move. Adding the six border states cuts the 1924 peak from 41.7 to 35.4 and the 1964 trough from −15.2 to −9.5. But the first negative year is **1964 either way**, and the permanent flip is **1984 either way**.

What a good answer says

3/4: Reports that the numbers changed but the shape didn't, and identifies 1964 as stable.

4/4: Draws the right inference — this is a **robustness check the finding passed**. The realignment is not an artifact of a convenient state list. A student who says “I tried to break it and couldn’t, which makes me believe it more” is doing science.

4/4, **best**: notices *why* the magnitudes shrink — the border states were always less Democratic than the Deep South, so adding them dilutes both extremes. The dilution is itself evidence: it says the effect is concentrated where the theory predicts.

Common stumble: treating “the numbers changed” as evidence the finding is shaky. Push back — the question is whether the *conclusion* changed, and it didn’t. This distinction is worth a minute of class time.

Option B — Pick a state and follow it

Verified output

```
for (s in c("MS", "VT")) {  
  x <- pres[pres$state_abbrev == s, ]  
  cat("\n", s, "\n"); print(x[x$year %in% c(1948, 1960, 1964, 1972, 2024),  
                             c("year", "d2", "winner")])  
}
```

```
#>  
#> MS  
#>   year      d2      winner  
#> 926  1948 79.38631 Strom Thurmond  
#> 1072 1960 59.56401 Harry S. Truman  
#> 1123 1964 12.86000 Barry Goldwater  
#> 1225 1972 20.06542 Richard Nixon  
#> 1888 2024 38.42653 Donald Trump  
#>  
#> VT  
#>   year      d2      winner  
#> 947  1948 37.49746 Thomas E. Dewey  
#> 1093 1960 41.35000 Richard Nixon  
#> 1144 1964 66.30663 Lyndon B. Johnson  
#> 1246 1972 36.79007 Richard Nixon  
#> 1909 2024 66.38586 Kamala Harris
```

Mississippi: 79.4 Democratic two-party share in 1948 → 12.9 in 1964 → 38.4 in 2024. **Vermont**: 37.5 in 1948 → 66.3 in 1964 → 66.4 in 2024.

They have swapped places almost exactly. 1964 was the **first time Vermont ever voted Democratic for president**.

What a good answer says

3/4: Narrates the line and identifies the break year for their state.

4/4: Notices that Mississippi 1948 shows d2 of 79.4 while the **winner** column says **Strom Thurmond**. Both are true: Truman took 79% of the Truman-versus-Dewey vote, and neither of them won the state,

because the Dixiecrat took it. That is the two-party-share limitation from Part 2 biting in the exact place the lab warned it would. Students who catch this have connected two parts of the lab unprompted.

4/4, best: answers the campaign question seriously. A 1950 strategist wrote Mississippi off as unwinnable-because-already-won and Vermont as unwinnable-because-already-lost; in 2024 a strategist writes both off for exactly the opposite reasons. The state changed completely and its *campaign treatment* did not change at all. Uncontested is uncontested.

Common stumble: picking a state that never did anything interesting (Utah, Nebraska) and concluding the lab is boring. Redirect them to MS, VT, WV, CA, or GA in office hours.

Option C — When did the battleground shrink?

Verified output

```
tot <- as.data.frame(table(pres$year)); names(tot) <- c("year","n")
tot$year <- as.numeric(as.character(tot$year))
for (mg in c(2, 5, 10)) {
  pres$c1 <- abs(pres$d2 - 50) < mg
  n <- aggregate(c1 ~ year, data = pres, FUN = sum, na.rm = TRUE)
  n <- merge(n, tot, by = "year"); n$pct <- round(100 * n$c1 / n$n, 1)
  cat("margin", mg, ":",
      paste0(n$year[n$year %in% c(1960,1976,1992,2008,2024)], "=",
            n$pct[n$year %in% c(1960,1976,1992,2008,2024)], "%", collapse = "  "), "\n")
}
```

```
#> margin 2 : 1960=36% 1976=33.3% 1992=19.6% 2008=9.8% 2024=13.7%
#> margin 5 : 1960=66% 1976=60.8% 1992=52.9% 2008=29.4% 2024=25.5%
#> margin 10 : 1960=88% 1976=86.3% 1992=88.2% 2008=64.7% 2024=58.8%
```

threshold	1960	1976	1992	2008	2024
±2 pts	36.0%	33.3%	19.6%	9.8%	13.7%
±5 pts	66.0%	60.8%	52.9%	29.4%	25.5%
±10 pts	88.0%	86.3%	88.2%	64.7%	58.8%

What a good answer says

3/4: Confirms the decline holds as a percentage, not just a count.

4/4: Reports honestly that the finding is **robust at ±5 and ±10 but wobbles at ±2**, where 2024 (13.7%) is *higher* than 2008 (9.8%). At the tightest cut-off you are counting a handful of states and the series gets noisy. The right conclusion is “the decline is real and large, and the tightest threshold is too noisy to add anything” — not “the finding failed.”

4/4, best: notices the ±10 row barely moves across 1960, 1976 and 1992 and then drops — and then goes and looks at the years the table does not print, where it is not flat at all. It falls to 25.5% in 1972 and 37.3% in 1984, both national landslides, and climbs back to 88.2% in 1992. At ±10 the measure is picking up **how lopsided the national result was** as much as how many states were genuinely in play, and the

three early columns in the table all happen to be close national elections. Compare like with like — close elections only — and the decline survives and is still late: 88.2% in 1992 against 58.8% in 2024, both close nationally. That is a genuinely open question and good Option D material.

Do not award the 4 for “the ± 10 row is flat until 2008” on its own. It reads that way only because of which five years the table prints; across every election in the file the ± 10 share spans 67.4 points. This is the same lesson as Option A in reverse — there, sampling the state list did not change the conclusion; here, sampling the years does.

Common stumble: reporting only ± 5 because it worked and not mentioning the others. Ask for all three in feedback.

Option D — Their own question

Reward curiosity over machinery. The 1960 Truman rows, the 1912 Roosevelt states, and the Vermont–Mississippi swap are all live threads.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **This lab rewards noticing.** The strongest submissions will flag something the lab did not point at — an odd year, a state that misbehaves, a label that cannot be right. Say so in feedback so the habit spreads.
- **Reward robustness reasoning.** Options A and C are both robustness checks in disguise. A student who distinguishes “the number moved” from “the conclusion moved” has learned the most transferable thing here.

Anticipated questions

- “*Why does the file say Truman won states in 1960?*” — Alabama and Mississippi ran **unpledged elector** slates; Mississippi’s won. The compiler needed a name and used a Democrat’s. Covered in Part 3, and it is the best single example in the course of a data error that leaves the numbers intact and the labels wrong.
- “*Why is the South missing in 1864?*” — Eleven states were in rebellion. The file records this only as absent rows.
- “*Are the 19th-century numbers trustworthy?*” — Partly. The bigger problem is not accuracy but meaning: post-1877 Southern returns are accurate counts of elections conducted under systematic disenfranchisement. Say this out loud; it connects back to **Session 3 (§203)** and forward to the **disenfranchisement/** lab.
- “*Isn’t two-party share cheating?*” — It is a choice with a cost, stated in Part 2. 1912, 1968 and 1992 are where it hurts.

Connections forward

- **campaign-visits** (retired, on disk) is the direct sequel: 13 competitive states is *why* candidates go where they go. Refer back to Part 5 by name.
- **Session 12 (roll calls and the bench)** — the realignment here is the mass-level counterpart to the elite polarization students will measure there. Same phenomenon, two instruments.
- **Session 3 (the ACS and §203)**, which ran last week — Part 3's disenfranchisement caveat is its historical setup. The Solid South was solid partly because most Black Southerners could not vote, and the 1965 Act is what changed the electorate underneath these lines.
- **Session 1 (election returns)** — the Truman rows and the sparse **other** column are the same provenance lesson, now in a historical file.